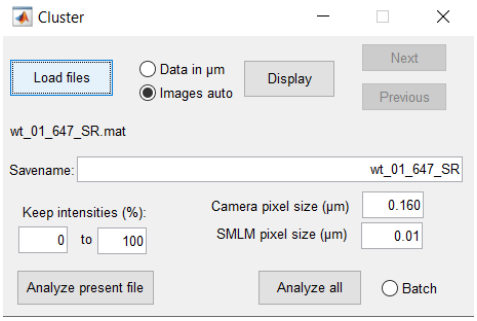
Cluster

This window is the main menu for clustering analysis. Use **Load files** to select the .mat files containing detections.

Please note:

- Click on **Data in µm** if your detection data is already converted to µm, before loading the files.
- Cluster.m requires files named *name*.mat (detections) and *name*.rnd (ROIs). You can create .rnd files using **Segment.m**.



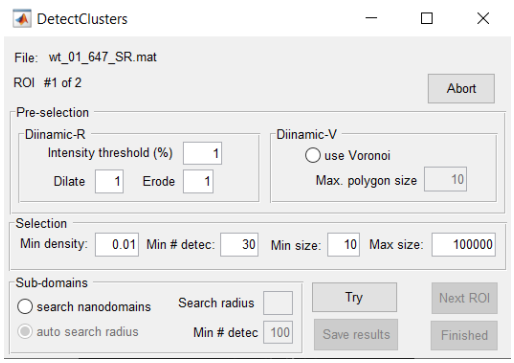
If **Images auto** is selected, the pointillistic and rendered images will be displayed automatically for the active file. Otherwise, you can use **Display** to visualize them. If you loaded a list of files, use **Next** or **Previous** to move to the next or previous file, respectively.

Keep intensities options may help the visualization eliminating detections with intensities below or above the minimum and maximum values provided (in %), but all the detections will be considered for the analysis.

Analyze present file will, as it says, perform the analysis only on the present file. If you loaded a list of files, **Analyze all** will sequentially analyze all the files. In these two cases, the clustering detection will ask for confirmation at each step and you will be able to tests different parameters on the same file. When you select **Batch** instead, **Analyze all** will ask to set the parameters for the first file and then it will automatically analyze all the files with the same parameters.

The analysis is done only on ROIs. If you want to analyze the whole image at once, you must select a ROI covering the entire image. This can be done also with Segment.m.

Clicking on one of the Analysis buttons will open a new window, DetectClusters, where you can choose the strategy to select candidate clusters (Pre-selection), enter parameters for pre-selection and selection of clusters, and chose to perform the detection of subdomains.

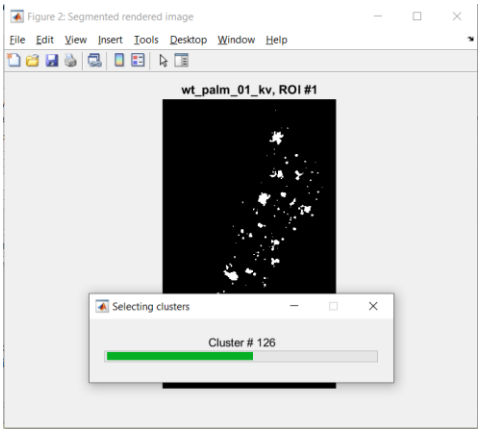


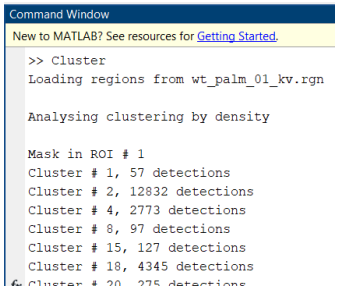
<u>Diinamic-R</u> is the option by default. It uses the rendered image to create a mask with the **Intensity threshold**. You may improve the mask using the morphological operators **Dilate** and **Erode** (enter the number of pixels to add or remove).

The selection of clusters is done considering:

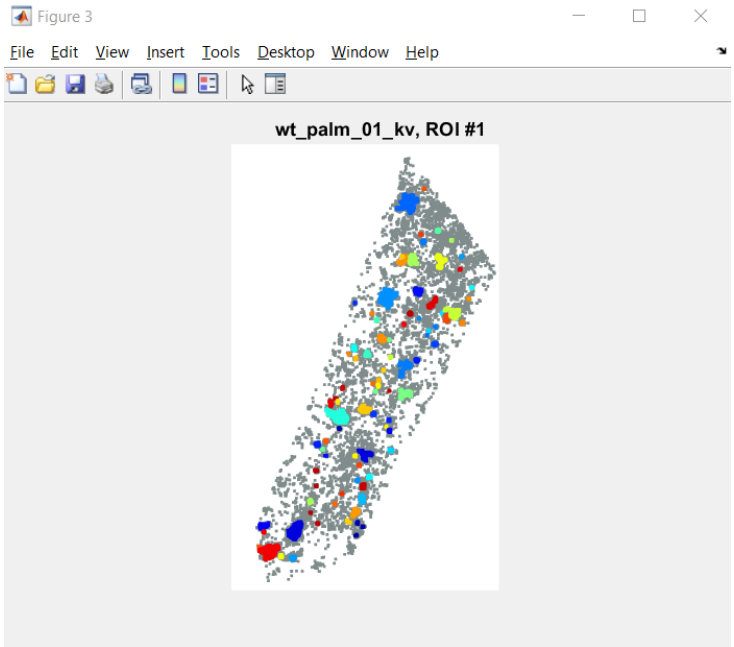
- Minimum density (**Min density**) per pixel (SMLM pixel),
- Minimum number of detections in the cluster (**Min # detec**),
- A minimum (**Min size**) and a maximum (**Max. size**) cluster size (number of pixels).

Try will launch the analysis on each ROI of the file. The rendered mask is shown on an independent figure, and you can follow the progression of the analysis on the Command window (the number of detections in *<u>selected</u>* clusters is shown). The wait bar also shows the progression. Close it to abort the analysis. If you want to stop the loop over a list of files, close the window or use the **Abort** button. Please note that partial results are not saved.



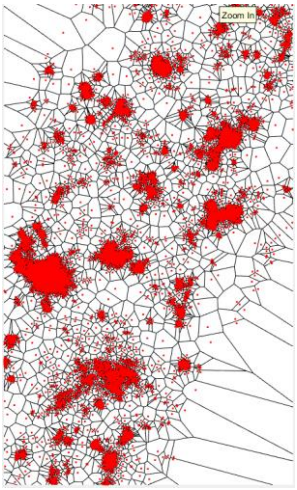


A new figure shows the selection results for the ROI, with the detections belonging to clusters appearing in color (each cluster has a different color) and those non-clustered detections appearing in gray.



<u>DiinamicV</u> uses Voronoi tessellation to create the mask. Select **use Voronoi** if you want to use this strategy, and enter the **Max. polygon size** (arbitrary units) to create the mask. The result will include

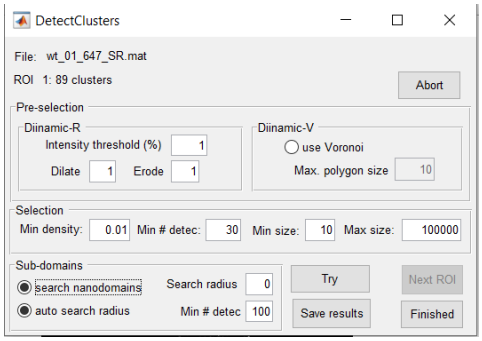
detections whose polygons are smaller than this threshold. An image with the Voronoi tessellation is displayed. The selection of clusters is done as for DiinamicR.



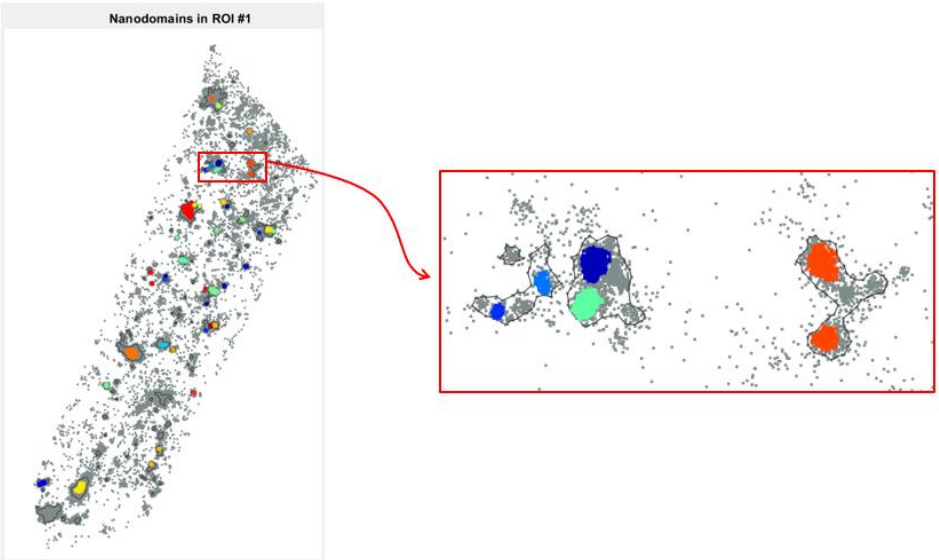
Exemple of Voronoi tessellation on SMLM data

Subdomains inside clusters

The detection of sub-domains on the selected clusters is done if you click on **search nanodomains**.



This part uses a DBSCAN algorithm: it needs a **search radius** and the minimum number of detections in the subdomain (**Min # detect**). The calculation of the search radius can be done automatically (**auto search radius**), based on the mean distance between detections in the sample.



Clusters that were detected previously are shown by a grey contour line that depicts their border. Nanoclusters appear in color. Note that detections themselves are not represented in scale so their size change if you zoom in or zoom out. The number of nanoclusters (and clusters) found are shown on DetectClusters window.

Results

If you agree with the results, click on **Save results** to keep them. Two files will be saved: *name*-roidata_*number of the ROI*.mat and *name*-selection.tif.

If there are more than one ROI on the data file, use **Next ROI** to move to the next ROI and click on **Try** to launch the detection on this new ROI.

When all the ROIs were analyzed, click on **Finished**. A new results file will be generated: *name*-allclust.txt. If you also made the detection of subdomains, these results will be saved in another file called *name*-nanoclust.txt. In addition to the results, there will be a text file that recapitulates the analysis that was done (Parameters_clustering_*date of analysis*.txt):

```
Parameters for clustering analysis

Analysis of density after segmentation with rendered image. Intensity threshold: 1%.
Segmented image improved with dilation: 1 px and erosion: 1 px.

Cluster selection criteria:
Minimum density: 0.01. Minimum number of detections: 30
Minimum cluster size (nm): 10. Maximum cluster size (nm): 100000

No nanodomain detection

              Files:
    wt_01_647_SR.mat
    wt_03_647_SR.mat
```

Example after analysis with subdomain detection:

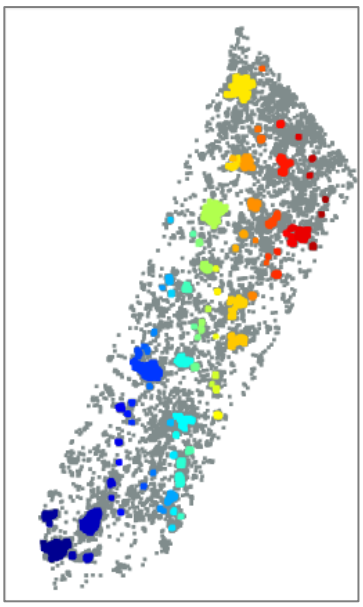
```
Parameters for clustering analysis

Analysis of density after segmentation with rendered image. Intensity threshold: 1%.
Segmented image improved with dilation: 1 px and erosion: 1 px.

Cluster selection criteria:
Minimum density: 0.01. Minimum number of detections: 30
Minimum cluster size (nm): 10. Maximum cluster size (nm): 100000

Including nanodomain detection by DBSCAN. Mninimum number of detections in the nanodomain : 100
Search radius calculated automatically : 5.95
              Files:
    wt_01_647_SR.mat
```

Format of results

Files *name*-allclust.txt: groups the results of all the ROI of a data file.

ROI #	cluster #	number of detections	detections in ROI	Cluster area	Cluster density
1.0000000e+00	5.0000000e+00	6.8000000e+01	3.7999000e+04	6.2571744e-03	1.0867525e+04
1.0000000e+00	6.0000000e+00	8.3000000e+01	3.7999000e+04	8.1745284e-03	1.0153491e+04
1.0000000e+00	1.5000000e+01	3.2000000e+01	3.7999000e+04	2.0911326e-03	1.5302712e+04

Files *name*-nanoclust.txt: groups the results of all the ROI of a data file.

ROI #	cluster #	nanodomain #	number of detections in the nanodomain	number of detections in the cluster	nanodomain area	nanodomain density

Files *name*-selection.tif: Image of the selected clusters (each cluster appears in a different color).